

Mainul Islam

Research Experience, Skills, and Interests

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My background sits at the intersection of embedded control systems, robotics, and applied machine learning. Across five research projects and two years of hands-on robotics engineering, I have moved from designing the low-level control and communication backbone of multi-node robotic systems to validating control algorithms in simulation and applying deep learning to real-world perception problems. I am seeking a Ph.D. position to formalize this experience into rigorous research on AI-enabled digital twins and control co-design for autonomous systems.

Control Systems, Robotics & Simulation

- **MonoSIM (published, IEEE):** Co-developed an open-source Software-in-the-Loop framework for Ackermann-steered vehicles, built to validate control algorithms before hardware deployment. Implemented and compared Model Predictive Control (MPC) and PID controllers against a monocular-vision lane-detection pipeline — direct experience with the simulate-design-validate loop central to control co-design and digital twins.
- **Multi-node Real-Time Control (published, IEEE ICCIT):** Designed and timed a master-slave RS485/UART communication architecture across 11+ STM32 microcontrollers coordinating a 6-DOF robotic arm and drive system, using DMA and CRC for reliable, low-latency feedback — foundational experience in the real-time data pipelines that AI-enabled digital twins depend on.
- **Teleoperated Robotic Manipulator ("Manportable"):** Built the system architecture for an 11-motor, dual-CAN-bus robot with a custom inverse-kinematics solver for a 7-DOF arm, ROS2-based integration with full sensor feedback, and fault/failsafe logic — practical exposure to the design-control interaction that motivates control co-design.
- **Autonomous Ground Vehicle:** Contributed an Extended Kalman Filter built from scratch to fuse dual-camera AprilTag detections with wheel odometry, and helped build the ROS2 stack for localization, path planning, and LIDAR-based obstacle avoidance — core state-estimation skills for model-based and learning-based control.

Machine Learning & Deep Learning

- **Bangla License Plate Recognition (published, ICARA):** Designed a two-stage adaptive YOLOv8 training strategy with phase-aware augmentation for detection, paired with a ViT + BanglaBERT OCR pipeline, validated on real low-light CCTV data — hands-on deep learning model design and rigorous benchmarking.
- **Solar Panel Fault Detection (published, Solar Energy):** Contributed to a hybrid RefineU-Net segmentation and MobileNetV3 classification framework for multi-class fault detection, incorporating explainable AI (LIME) for interpretability — experience with efficient architectures and trustworthy-model considerations relevant to AI-driven decision-making.
- **Long-Term Energy Demand Forecasting (published, IEEE):** Benchmarked Random Forest, KNN, XGBoost, and LightGBM on a million-sample utility dataset for long-term demand prediction — direct experience with data-driven modeling and comparative optimization of learning algorithms.

Technical Skills

Python (PyTorch, scikit-learn), C/C++ (embedded, STM32), ROS2, real-time embedded control (UART, RS485, CAN bus, Modbus, DMA), classical estimation and control (EKF, PID, MPC), PCB design, Git, and end-to-end deployment of vision and forecasting models from data collection through publication.

Research Interests

My experience validating controllers in simulation before hardware deployment (MonoSIM) and building the real-time sensing/communication backbone for multi-agent robots naturally extends toward AI-enabled digital twins: systems that continuously learn from live data to inform design and control decisions. I am especially drawn to control co-design that couples reinforcement learning with physical design optimization, and to multi-fidelity, simulation-grounded approaches for dynamic autonomous and robotic systems and I am eager to bring my combined control, embedded-systems, and deep-learning background to that work.